

PART A — MODULE 1 QUESTIONS

MATH10-M1-001

Prompt:

What is the solution to $(3x)/5 + 4 = 13$?

- A) 9
- B) 12
- C) 15
- D) 20

MATH10-M1-002

Prompt:

A car travels 72 miles using 3 gallons of gas. At this rate, how many miles can the car travel using 8 gallons of gas?

- A) 144
- B) 168
- C) 192
- D) 216

MATH10-M1-003

Prompt:

The table shows the numbers of hours that students spent volunteering in one month.

Hours volunteered: 0, 1, 2, 3

Number of students: 2, 5, 9, 4

How many students volunteered at least 1 hour?

- A) 5
- B) 9
- C) 13
- D) 18

MATH10-M1-004

Prompt:

A linear function f has $f(1) = 8$ and $f(4) = 20$. What is $f(2)$?

- A) 10
- B) 12
- C) 14
- D) 16

MATH10-M1-005

Prompt:

A student needs to read more than 120 pages this week. The student has already read 46 pages and will read p more pages. Which inequality represents this situation?

- A) $p + 46 > 120$
- B) $p + 46 < 120$
- C) $46p > 120$
- D) $p - 46 > 120$

MATH10-M1-006

Prompt:

What is the value of x in the solution to the system of equations?

$$y = x + 5$$

$$2x + y = 26$$

- A) 5
- B) 7
- C) 9
- D) 12

MATH10-M1-007

Prompt:

Which expression is equivalent to $4(2a - 1) + 3(a + 6)$?

- A) $11a + 14$
- B) $11a + 17$
- C) $8a + 14$
- D) $8a + 17$

MATH10-M1-008

Prompt:

Which expression is equivalent to $(d^3)(d^4) \div d^2$, where $d \neq 0$?

- A) d^5
- B) d^6
- C) d^7
- D) d^9

MATH10-M1-009

Prompt:

A bag contains 5 orange markers, 7 purple markers, and 8 black markers. If one marker is selected at random, what is the probability that the marker selected is orange or purple?

- A) $\frac{1}{4}$
- B) $\frac{3}{5}$
- C) $\frac{7}{10}$
- D) $\frac{4}{5}$

MATH10-M1-010

Prompt:

A regular pentagon has side length 9 centimeters. What is the perimeter, in centimeters, of the pentagon?

- A) 14
- B) 36
- C) 45
- D) 81

MATH10-M1-011

Prompt:

In the xy -plane, point A has coordinates $(-6, 3)$. Point B is 8 units to the right of point A and has the same y -coordinate. What are the coordinates of point B?

- A) $(-14, 3)$
- B) $(2, 3)$
- C) $(-6, 11)$
- D) $(8, 3)$

MATH10-M1-012

Prompt:

What is the value of $|-9| + \sqrt{49}$?

- A) 2
- B) 14
- C) 16
- D) 58

MATH10-M1-013

Prompt:

If $g(x) = x^2 + 3x$, what is $g(-2)$?

- A) -10
- B) -2
- C) 2
- D) 10

MATH10-M1-014

Prompt:

The equation $x^2 - 25 = 0$ has two solutions. What is the greater solution?

- A) -25
- B) -5
- C) 5
- D) 25

MATH10-M1-015

Prompt:

A store sold 36 water bottles on Monday and 45 water bottles on Tuesday. The number of water bottles sold increased by what fraction of Monday's number?

- A) $\frac{1}{5}$
- B) $\frac{1}{4}$
- C) $\frac{1}{3}$
- D) $\frac{5}{4}$

MATH10-M1-016

Prompt:

The table shows the total cost of buying different numbers of identical folders.

Folders: 3, 6, 9, 12

Total cost in dollars: 4.50, 9.00, 13.50, 18.00

What is the cost, in dollars, of 1 folder?

- A) 1.25
- B) 1.50
- C) 2.00
- D) 3.00

MATH10-M1-017

Prompt:

A right triangle has one leg of length 7 and a hypotenuse of length 25. What is the length of the other leg?

- A) 18

- B) 20
- C) 24
- D) 32

MATH10-M1-018

Prompt:

The formula $V = s^3$ gives the volume V of a cube with side length s . What is the volume of a cube with side length 5 inches?

- A) 15
- B) 25
- C) 75
- D) 125

MATH10-M1-019

Prompt:

The equation $M = 12 + 0.75n$ models the mass M , in grams, of a container holding n identical beads. What does 0.75 represent in this equation?

- A) The mass, in grams, of the empty container
- B) The mass, in grams, of each bead
- C) The number of beads in the container
- D) The total mass, in grams, of 0.75 containers

MATH10-M1-020

Prompt:

A set of 10 cards contains 4 blue cards and 6 green cards. What percent of the cards are blue?

- A) 4%
- B) 10%
- C) 40%
- D) 60%

MATH10-M1-021

Prompt:

The values in a data set are 3, 7, 8, 10, and 12. What is the median of the data set?

- A) 7
- B) 8
- C) 9
- D) 10

MATH10-M1-022

Prompt:

A rectangle has area 72 square meters and length 9 meters. What is the width, in meters, of the rectangle?

- A) 6
- B) 8
- C) 9
- D) 18

PART B — MODULE 2 QUESTIONS

MATH10-M2-001

Prompt:

What is the solution to $6x - 2(4x - 9) = 30$?

- A) -6
- B) -4
- C) 4
- D) 6

MATH10-M2-002

Prompt:

The system of equations has solution (x, y) .

$$2x + 3y = 28$$

$$4x - y = 14$$

What is the value of $x - y$?

- A) -2
- B) 1
- C) 4
- D) 7

MATH10-M2-003

Prompt:

The function f is defined by $f(x) = x^2 + 6x + 4$. What is the minimum value of $f(x)$?

- A) -9
- B) -5
- C) 4
- D) 13

MATH10-M2-004

Prompt:

For $x \neq 2$, which expression is equivalent to $(x^2 - x - 2)/(x - 2)$?

- A) $x - 2$
- B) $x - 1$
- C) $x + 1$
- D) $x + 2$

MATH10-M2-005

Prompt:

A population of fish in a pond is modeled by $P(t) = 900(1.08)^t$, where t is the number of years after the population was first measured. Which statement is the best interpretation of the number 1.08 in this model?

- A) The population starts at 1.08 fish.
- B) The population increases by 8% each year.
- C) The population increases by 108 fish each year.
- D) The population is multiplied by 900 each year.

MATH10-M2-006

Prompt:

A camera costs \$180 after a 20% discount. What was the original price of the camera?

- A) \$200
- B) \$216
- C) \$225
- D) \$240

MATH10-M2-007

Prompt:

The table shows values of a linear function f .

x : $-4, -1, 2, 5$

$f(x)$: $19, 10, 1, -8$

Which equation defines f ?

A) $f(x) = -3x + 7$

B) $f(x) = -2x + 11$

C) $f(x) = 3x + 7$

D) $f(x) = 4x - 3$

MATH10-M2-008

Prompt:

A rectangular sandbox is 10 feet long and 6 feet wide. Sand is poured into the sandbox to a uniform depth of 0.5 foot. What is the volume, in cubic feet, of sand in the sandbox?

A) 16

B) 30

C) 60

D) 120

MATH10-M2-009

Prompt:

In a right triangle, $\sin \theta = 12/13$. What is $\tan \theta$?

A) $5/13$

B) $5/12$

C) $12/5$

D) $13/5$

MATH10-M2-010

Prompt:

A line passes through the points $(-3, -2)$ and $(5, 10)$. What is the y -intercept of the line?

A) $-5/2$

B) 1

C) $5/2$

D) 4

MATH10-M2-011

Prompt:

The functions f and g are defined by $f(x) = x^2 - 3$ and $g(x) = 2x + 1$. What is $f(g(2))$?

A) 7

B) 16

C) 22

D) 25

MATH10-M2-012

Prompt:

What is the positive solution to $|3x - 4| = 11$?

A) $5/3$

B) $7/3$

- C) 5
- D) 15

MATH10-M2-013

Prompt:

In the xy -plane, a circle has center $(1, -4)$ and radius 3. Which point lies on the circle?

- A) $(1, -1)$
- B) $(4, 0)$
- C) $(-2, -8)$
- D) $(3, -4)$

MATH10-M2-014

Prompt:

For what value of b does the equation $7x + 5 = bx + 5$ have infinitely many solutions?

- A) 0
- B) 5
- C) 7
- D) 12

MATH10-M2-015

Prompt:

The table shows values of a quadratic function q .

x : $-2, -1, 0, 1$
 $q(x)$: $13, 4, -1, -2$

Which equation could define q ?

- A) $q(x) = 2x^2 - 3x - 1$
- B) $q(x) = 2x^2 + 3x - 1$
- C) $q(x) = -2x^2 - 3x - 1$
- D) $q(x) = -2x^2 + 3x - 1$

MATH10-M2-016

Prompt:

Which expression is equivalent to $(10x^3y^2 - 15x^2y)/(5x^2y)$, where $x \neq 0$ and $y \neq 0$?

- A) $2xy - 3$
- B) $2xy - 15$
- C) $5x - 3y$
- D) $5xy - 3$

MATH10-M2-017

Prompt:

A class has 24 students. Of the students, 15 are taking biology, 11 are taking chemistry, and 6 are taking both biology and chemistry. How many students are taking biology or chemistry, but not both?

- A) 8
- B) 14
- C) 20
- D) 22

MATH10-M2-018

Prompt:

A cone has radius 6 centimeters and height 10 centimeters. The volume of a cone is $V = (1/3)\pi r^2 h$. What is the volume, in cubic centimeters, of the cone?

- A) 60π
- B) 120π
- C) 180π
- D) 360π

MATH10-M2-019

Prompt:

For all values of x , $(x + 4)^2 - k = x^2 + 8x + 7$. What is the value of k ?

- A) 7
- B) 9
- C) 16
- D) 23

MATH10-M2-020

Prompt:

A data set has 8 values with a mean of 25. A value of 41 is added to the data set. What is the mean of the new data set?

- A) 26
- B) 26.5
- C) 27
- D) 27.5

MATH10-M2-021

Prompt:

A school sold adult tickets for \$10 each and child tickets for \$6 each. A total of 58 tickets were sold for \$460. How many child tickets were sold?

- A) 20
- B) 25
- C) 30
- D) 35

MATH10-M2-022

Prompt:

The function h is defined by $h(x) = a(x - 5)^2 - 2$, where a is a constant. If $h(1) = 30$, what is $h(7)$?

- A) 6
- B) 8
- C) 10
- D) 14

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH10-M1-001 — Correct: 15 — Explanation: Subtract 4 from both sides to get $(3x)/5 = 9$. Multiply both sides by 5 to get $3x = 45$, so $x = 15$.

MATH10-M1-002 — Correct: 192 — Explanation: The car travels $72 \div 3 = 24$ miles per gallon. Using 8 gallons, it can travel $24 \times 8 = 192$ miles.

MATH10-M1-003 — Correct: 18 — Explanation: Students who volunteered at least 1 hour are those in the 1-hour, 2-hour, and 3-hour groups: $5 + 9 + 4 = 18$.

MATH10-M1-004 — Correct: 12 — Explanation: The rate of change is $(20 - 8)/(4 - 1) = 4$. From $x = 1$ to $x = 2$, f increases by 4, so $f(2) = 12$.

MATH10-M1-005 — Correct: $p + 46 > 120$ — Explanation: The total pages read will be $46 + p$, and this total must be more than 120.

MATH10-M1-006 — Correct: 7 — Explanation: Substitute $y = x + 5$ into $2x + y = 26$ to get $2x + x + 5 = 26$. Then $3x = 21$, so $x = 7$.

MATH10-M1-007 — Correct: $11a + 14$ — Explanation: Distribute and combine like terms: $4(2a - 1) + 3(a + 6) = 8a - 4 + 3a + 18 = 11a + 14$.

MATH10-M1-008 — Correct: d^5 — Explanation: Add exponents when multiplying and subtract when dividing: $d^3 \times d^4 \div d^2 = d^7 \div d^2 = d^5$.

MATH10-M1-009 — Correct: $3/5$ — Explanation: There are 20 markers total, and $5 + 7 = 12$ are orange or purple. The probability is $12/20 = 3/5$.

MATH10-M1-010 — Correct: 45 — Explanation: A regular pentagon has 5 equal sides, so the perimeter is $5 \times 9 = 45$ centimeters.

MATH10-M1-011 — Correct: (2, 3) — Explanation: Moving 8 units to the right increases the x -coordinate from -6 to 2 , while the y -coordinate remains 3 .

MATH10-M1-012 — Correct: 16 — Explanation: $|-9| = 9$ and $\sqrt{49} = 7$, so the sum is 16.

MATH10-M1-013 — Correct: -2 — Explanation: Substitute -2 for x : $g(-2) = (-2)^2 + 3(-2) = 4 - 6 = -2$.

MATH10-M1-014 — Correct: 5 — Explanation: $x^2 - 25 = 0$ gives $x^2 = 25$, so $x = -5$ or $x = 5$. The greater solution is 5.

MATH10-M1-015 — Correct: $1/4$ — Explanation: The increase was $45 - 36 = 9$. As a fraction of Monday's number, this is $9/36 = 1/4$.

MATH10-M1-016 — Correct: 1.50 — Explanation: The cost per folder is $4.50 \div 3 = 1.50$ dollars.

MATH10-M1-017 — Correct: 24 — Explanation: By the Pythagorean theorem, the other leg has length $\sqrt{(25^2 - 7^2)} = \sqrt{576} = 24$.

MATH10-M1-018 — Correct: 125 — Explanation: Substitute $s = 5$ into $V = s^3$ to get $V = 5^3 = 125$.

MATH10-M1-019 — Correct: The mass, in grams, of each bead — Explanation: In $M = 12 + 0.75n$, the coefficient 0.75 represents the increase in mass for each additional bead.

MATH10-M1-020 — Correct: 40% — Explanation: There are 4 blue cards out of 10 cards, so $4/10 = 0.40$, or 40%.

MATH10-M1-021 — Correct: 8 — Explanation: The values are ordered, and the middle value is 8, so the median is 8.

MATH10-M1-022 — Correct: 8 — Explanation: Area = length $\times$ width, so $72 = 9w$. Thus $w = 8$.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH10-M2-001 — Correct: -6 — Explanation: Expand to get $6x - 8x + 18 = 30$. Then $-2x = 12$, so $x = -6$.

MATH10-M2-002 — Correct: 1 — Explanation: From $4x - y = 14$, $y = 4x - 14$. Substitute into $2x + 3y = 28$ to get $2x + 3(4x - 14) = 28$. Then $14x = 70$, so $x = 5$ and $y = 6$. Thus $x - y = -1$. The correct option should be -1 , so this item needs revision.

MATH10-M2-003 — Correct: -5 — Explanation: Complete the square: $x^2 + 6x + 4 = (x + 3)^2 - 5$. Since the squared term is nonnegative, the minimum value is -5 .

MATH10-M2-004 — Correct: $x + 1$ — Explanation: Factor the numerator: $x^2 - x - 2 = (x - 2)(x + 1)$. Since $x \neq 2$, the expression simplifies to $x + 1$.

MATH10-M2-005 — Correct: The population increases by 8% each year. — Explanation: Multiplying by 1.08 each year means the population is 108% of the previous year's population, an 8% increase.

MATH10-M2-006 — Correct: \$225 — Explanation: A 20% discount means the sale price is 80% of the original price. If $0.80p = 180$, then $p = 225$.

MATH10-M2-007 — Correct: $f(x) = -3x + 7$ — Explanation: The function decreases by 9 when x increases by 3, so the slope is -3 . Since $f(-1) = 10$, the equation $f(x) = -3x + 7$ matches the table.

MATH10-M2-008 — Correct: 30 — Explanation: The volume is length $\times$ width $\times$ depth: $10 \times 6 \times 0.5 = 30$ cubic feet.

MATH10-M2-009 — Correct: $12/5$ — Explanation: If $\sin \theta = 12/13$, the opposite side can be 12 and the hypotenuse 13. The adjacent side is 5, so $\tan \theta = 12/5$.

MATH10-M2-010 — Correct: $5/2$ — Explanation: The slope is $(10 - (-2))/(5 - (-3)) = 12/8 = 3/2$. Using $y = (3/2)x + b$ and point $(5, 10)$, $10 = 15/2 + b$, so $b = 5/2$.

MATH10-M2-011 — Correct: 22 — Explanation: First find $g(2) = 2(2) + 1 = 5$. Then $f(5) = 5^2 - 3 = 22$.

MATH10-M2-012 — Correct: 5 — Explanation: $|3x - 4| = 11$ gives $3x - 4 = 11$ or $3x - 4 = -11$. The positive solution is $x = 5$.

MATH10-M2-013 — Correct: $(1, -1)$ — Explanation: The point $(1, -1)$ is 3 units vertically from the center $(1, -4)$, so it lies on the circle.

MATH10-M2-014 — Correct: 7 — Explanation: For $7x + 5 = bx + 5$ to be true for all values of x , the coefficients of x must be equal, so $b = 7$.

MATH10-M2-015 — Correct: $q(x) = 2x^2 - 3x - 1$ — Explanation: Substituting $x = -2, -1, 0$, and 1 into $2x^2 - 3x - 1$ gives 13, 4, -1 , and -2 , matching the table.

MATH10-M2-016 — Correct: $2xy - 3$ — Explanation: Divide each term by $5x^2y$: $(10x^3y^2)/(5x^2y) - (15x^2y)/(5x^2y) = 2xy - 3$.

MATH10-M2-017 — Correct: 14 — Explanation: Biology only is $15 - 6 = 9$, and chemistry only is $11 - 6 = 5$. Thus the number taking biology or chemistry, but not both, is $9 + 5 = 14$.

MATH10-M2-018 — Correct: 120π — Explanation: Substitute $r = 6$ and $h = 10$ into $V = (1/3)\pi r^2 h$ to get $V = (1/3)\pi(36)(10) = 120\pi$.

MATH10-M2-019 — Correct: 9 — Explanation: Expand $(x + 4)^2 - k$ to get $x^2 + 8x + 16 - k$. Since this equals $x^2 + 8x + 7$, $16 - k = 7$, so $k = 9$.

MATH10-M2-020 — Correct: 26.5 — Explanation: The original sum is $8 \times 25 = 200$. Adding 41 gives 241 for 9 values, and $241 \div 9$ is about 26.78, so no listed choice is exact.

MATH10-M2-021 — Correct: 30 — Explanation: Let c be the number of child tickets. Then adult tickets are $58 - c$. The equation is $6c + 10(58 - c) = 460$. This gives $580 - 4c = 460$, so $c = 30$.

MATH10-M2-022 — Correct: 6 — Explanation: Substitute $x = 1$ into $h(x) = a(x - 5)^2 - 2$: $30 = 16a - 2$, so $32 = 16a$ and $a = 2$. Then $h(7) = 2(2^2) - 2 = 6$.